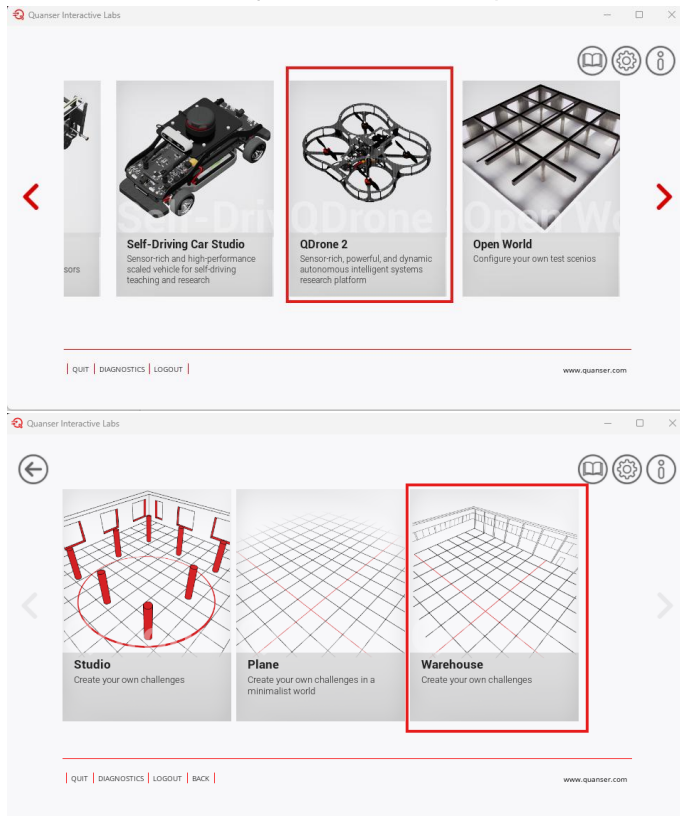


Virtual QDrone 2 Position Control Instructions

1. Open the Quanser Interactive Labs (QLabs) software.
2. Go to the QDrone 2 | Warehouse workspace.



3. Run the **spawn_drone2.py** Python script using the command “python spawn_qdrone2.py”.

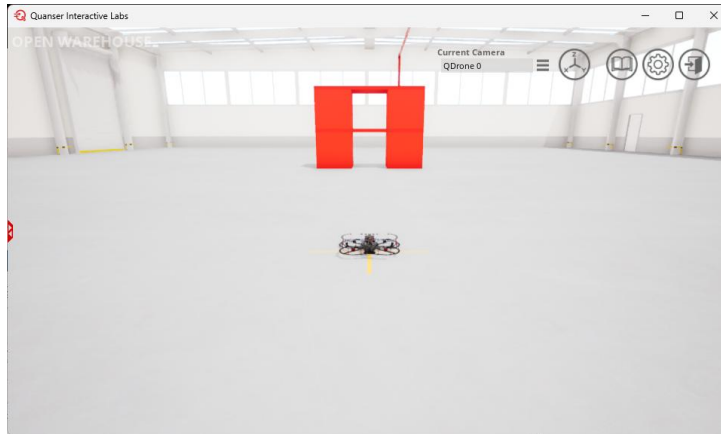
```
Windows PowerShell
PS C:\Users\mlevis\Documents\Quanser\5_research\autonomous_vehicles\qdrone2\virtual\position_control> ls

Directory: C:\Users\mlevis\Documents\Quanser\5_research\autonomous_vehicles\qdrone2\virtual\position_control

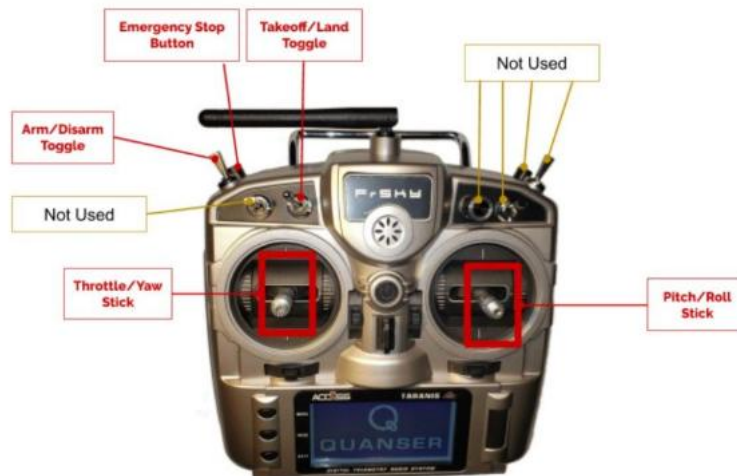
Mode                LastWriteTime         Length Name
----                -
d-----          2/11/2026   3:48 PM                scripts
-a-----          2/11/2026   3:48 PM             645 Motor_Mapping_7_Inch.m
-a-----          2/11/2026   3:48 PM            4543 spawn_qdrone2.py
-a-----          2/11/2026   3:48 PM       258584 virtual_Q02_DroneStack_PID_2024a.slx
-a-----          2/11/2026   3:48 PM       695774 virtual_Q02_MissionCtrl_2024a.slx

PS C:\Users\mlevis\Documents\Quanser\5_research\autonomous_vehicles\qdrone2\virtual\position_control> python .\spawn_qdrone2.py
Connecting to QLABS...
Connected
Done!
PS C:\Users\mlevis\Documents\Quanser\5_research\autonomous_vehicles\qdrone2\virtual\position_control> |
```

4. The Virtual QDrone 2 should be created in the workspace.

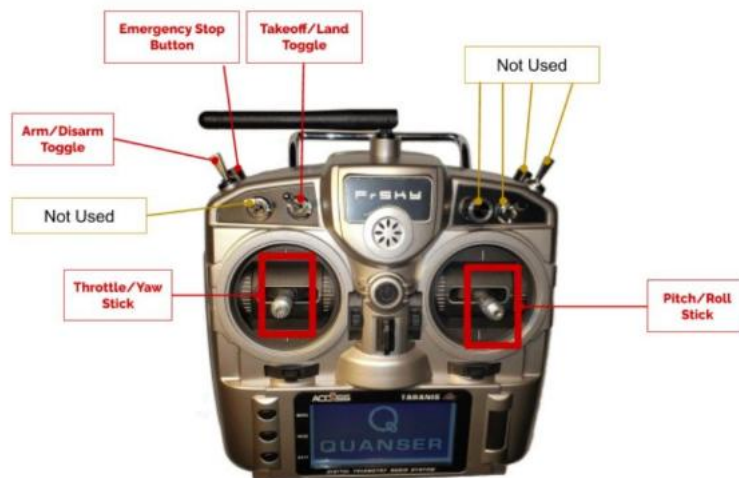


5. Open the *virtual_QD2_MissionCtrl_2024a* Simulink model in MATLAB if using the FrSky Simulator joystick. If you don't have this, open the *virtual_QD2_MissionCtrl_Keyboard_2024a* Simulink model to use the keyboard commands.
6. Click on Monitor & Tune to run it in QUARC.
7. Open a second MATLAB application.
8. Open the *virtual_QD2_DroneStack_PID_2024a* in Simulink in the second MATLAB.
9. Click on Monitor & Tune to run it in QUARC.
10. **Option 1:** If using the *virtual_QD2_MissionCtrl_2024a* Simulink model, use the joystick toggles to Arm and then Takeoff.



NOTE: Current release of joystick firmware should read "Quanser V2" which is compatible with the mapping in this model

Option 2: If using the *virtual_QD2_MissionCtrl_Keyboard_2024a* Simulink model, go to the Keyboard Command subsystem to Arm and Takeoff the virtual QDrone 2.



NOTE: Current release of joystick firmware should read "Quanser V2" which is compatible with the mapping in this model

16. **Option 2:** If using the virtual_QD2_MissionCtrl_Keyboard_2024a Simulink model, go to the Keyboard Command subsystem to *Land* and *Disarm*.
17. The QDrone 2 should land.